

AgiBot X2 Ground Recovery

Simulation, reinforcement learning and ROS 2 integration | 28 September 2026

Result: 5/5 supine recoveries. The selected PPO policy starts from random-weight training with every episode supine. All five 10-second validation episodes end in stable two-foot stance without other-body support. No upright-start pretraining or lift assistance is used in this lineage.

Robot and simulation

Item	Implementation
Model	Official AgiBot X2 Ultra v1.3.0 URDF, upstream revision 60c5de5; locally converted to USD.
Physics	Isaac Sim 6.0.1 / Isaac Lab 3.0.0-beta2.patch1 / PhysX; floating base, 200 x 200 m flat floor, self-collision.
Constraints	31 movable joints and 32 sensed bodies; vendor meshes, inertia, joint axes and actuator limits. Position commands use imported 98% soft limits.
Reset / rate	Supine pelvis height 0.190 m, neutral joints and zero velocity; collision clearance checked. Physics 200 Hz, policy 50 Hz; 10 s or 500 policy steps.

Fetch and import: `scripts/fetch_agibot_model.sh` and `scripts/import_x2_isaac.sh`. The generated USD stays local. Mesh collision shapes use the converter's convex-hull approximation.

Observation and action contract

Policy input	Count
Pelvis height; body-frame linear/angular velocity; projected gravity	1 + 3 + 3 + 3
Joint position / velocity	31 + 31
Left/right foot contact; whole-body ground-contact bits	2 + 32
Previous two policy actions	16
Total	122

Eight symmetric channels command hip pitch, knee, ankle pitch, shoulder pitch, elbow, waist pitch, ankle roll and hip roll. They vary 15 joint targets using $\text{centre} + \text{span} * \tanh(\text{action})$, then clamp to imported soft limits. Separate normalized actor/critic MLPs use 512/256/128 ELU layers.

The other 16 joints retain default targets and remain simulated and published: hip yaw; shoulder roll/yaw; wrist roll/pitch/yaw (both sides); waist roll/yaw; head pitch/yaw. Actions specify positions, not direct torques.

Evaluation seeds

One checkpoint, five reproducible supine starts. Each seed samples a different pose within the same limits; seed IDs are local choices.

Seed	Position variation	Orientation variation	Unchanged
101	X/Y: +/-4 cm Z: +/-1 mm	Roll/pitch: +/-0.005 rad Yaw: +/-0.02 rad	Neutral joints; zero velocity.
102	X/Y: +/-4 cm Z: +/-1 mm	Roll/pitch: +/-0.005 rad Yaw: +/-0.02 rad	Neutral joints; zero velocity.
103	X/Y: +/-4 cm Z: +/-1 mm	Roll/pitch: +/-0.005 rad Yaw: +/-0.02 rad	Neutral joints; zero velocity.
104	X/Y: +/-4 cm Z: +/-1 mm	Roll/pitch: +/-0.005 rad Yaw: +/-0.02 rad	Neutral joints; zero velocity.
105	X/Y: +/-4 cm Z: +/-1 mm	Roll/pitch: +/-0.005 rad Yaw: +/-0.02 rad	Neutral joints; zero velocity.

Reward design and PPO experiment

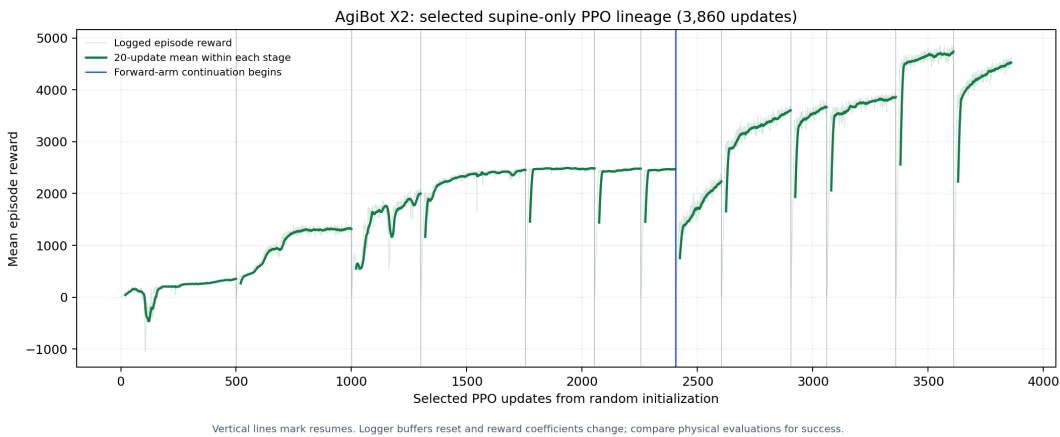
Signed height guides the rise without rewarding an inverted bridge. Transitional hand/body contacts remain allowed. Dense posture feedback and motion/command regularization guide the transition from rising to sustained foot-only stance.

Term	Weight	Purpose / implementation
Signed pelvis height	40	$\text{clip}(z/0.68) * \text{clip}((1-g_z)/2)$.
Head height / upright	5 / 5	Clipped exponential head height; $\exp(-g_z)$.
Feet / other support	10 / -1	Two-foot contact; discourage other support near standing.
Balance / strict stance	20 / 20	Low velocity near standing; complete physical success predicate.
Stance proximity	20	Exponential proximity to stance. Training widths: tilt 0.35, speed 0.8 m/s and angular speed 2 rad/s.
Leg pose	20	Neutral hip pitch/knee/ankle pitch; gated above 0.40 m and by upright orientation.
Forward arm pose	40	Supported upright shoulder -0.22 and elbow -1.17 rad; Gaussian variance 0.5.
Arm commands	200 / 200	Supported shoulder proximity to $\text{atanh}(0.14)$, variance 0.2; combined shoulder/elbow proximity, variance 2.
Near-stance motion	-2	Penalize squared base speed and 0.1 times squared angular speed near standing.
Command saturation	-0.5	Squared excess outside effective raw-action limits; accounts for tanh and imported soft-limit clipping.
Action change / joint speed	-0.02 / -0.0005	Squared successive raw-action changes / joint velocities.
Torque / limit / failure	-1e-6 / -1 / -10	Squared effort, soft-limit violation, non-timeout safety termination.

RewardManager sums weight x term x 0.02 s. HumanUP [1] motivates rise/control terms; HoST [2] motivates post-task posture; FRASA [3] motivates compact symmetric control and desired-pose proximity. Weights, gates, widths and raw-command penalties are X2 choices, not copied paper constants.

Training configuration

PPO [4]: clip 0.2, gamma 0.99, GAE lambda 0.95; 5 epochs, 4 minibatches, value coefficient 1, gradient clip 1, desired KL 0.01. Each stage uses 3,000 environments and 32 steps/environment. The selected all-supine ancestry contains 3,860 updates: 1,000 stance-learning updates, 1,404 intermediate supported-posture updates and 1,456 selected forward-arm updates. Resumes retain actor, critic and optimizer; final arm stages adjust shoulder/elbow exploration std.



Complete 3,860-update selected supine-only lineage; faint: logged reward, dark: 20-update mean within each stage. Vertical lines mark resumes. Logger buffers and reward coefficients change, so the five-episode physical evaluation establishes success.

Five-episode validation

A recovery requires every condition below for at least 0.5 continuous seconds. Evaluation continues for the complete episode; terminal samples are captured before automatic reset. Final stance duration is reported separately from a brief earlier success.

Check	Threshold
Height	Pelvis at least 0.58 m.
Orientation	$\text{norm}(\text{projected_gravity_xy}) \leq 0.15$ and $g_z \leq -0.98$.
Base motion	Linear speed ≤ 0.25 m/s; angular speed ≤ 0.35 rad/s.
Support	Each foot ≥ 15 N ground force; every other body < 15 N.

Measured outcome

Seed	Recovery	Strict stance at end	Two feet at height peak
101	Yes	8.90 s	Yes
102	Yes	8.88 s	Yes
103	Yes	8.90 s	Yes
104	Yes	8.90 s	Yes
105	Yes	9.00 s	Yes

5/5 successful recoveries. All five end on both feet without other-body support, with 8.88-9.00 s of uninterrupted strict stance at episode end. The forward-bent arm pose is a local design preference: terminal shoulder pitch is about -0.39 to -0.44 rad and elbow about -0.85 to -0.87 rad. Both feet contact the floor at the height peak in all five required episodes.

Source: `reports/relaxed_v4_evaluation.json`. Checkpoint: `reports/checkpoints/x2_supine_model3847.pt`.

Failure analysis and scope

The original supine-only 500-update run scored 0/5. It reached 0.763-0.783 m while airborne and never obtained strict-stance reward. Earlier successful weights had mixed-start pretraining, which supplied early balance experience; that lineage is separate from this result.

Extending the original 12-reward, random-weight supine run to 2,000 updates still scored 0/5: the robot repeatedly rose and fell without meeting stable stance. This is a recorded comparison, not an isolated reward ablation.

The new lineage retains every supine reset while adding dense stance/leg-pose feedback and motion/action regularization. The first 500 updates still scored 0/5; continuing the same preset reached 5/5 after 1,000 updates. This is a combined intervention and larger training budget, not an ablation proving one reward alone caused the improvement.

The intermediate neutral-arm preference was optional, not an HRS criterion. We continued its supported-rise policy with an X2-specific forward-arm objective. The fixed-reward 1,500-update alternative also recovered 5/5 but had both feet airborne at the seed-101 height peak; this selected model keeps foot contact at the peak in all five required episodes.

Scope: nominal flat-floor simulation with small reset perturbations. Terrain variation, broad pose recovery, sensor noise, latency and real-hardware performance were not validated. Development failures and exact configuration provenance remain in Git and `docs/development_history.md`.

ROS 2 interface and reproduction

The Python package builds with colcon and launches recovery and telemetry nodes together. A local Unix socket connects ROS 2 Humble (Python 3.10) to the Isaac policy server (Python 3.12). Joint positions come from the live simulator.

Interface	ROS 2 type / behavior
/x2/start_recovery	std_srvs/srv/Trigger: acceptance before execution; a second request while busy returns false.
/x2/recovery_status	std_msgs/msg/String: IDLE, RUNNING, SUCCEEDED or FAILED.
/x2/joint_states	sensor_msgs/msg/JointState: timestamps and 31 measured joint names/positions while running.

Telemetry logs status and left_knee_joint at 1 Hz. ROS ends after the physical 0.5 s success hold; the separate full-length evaluation verifies final stance. Timeout is configurable. A fresh isolated colcon build passed; the selected exported policy was tested through the real ROS-Isaac connection.

Recorded check	Observed outcome
CLI recovery	RUNNING to SUCCEEDED after 81 simulation steps; timestamped 31-joint telemetry.
Concurrent request	First accepted; second returned false: Recovery already running.
Timeout 0.2 s	RUNNING to FAILED after 10 steps with timeout 0.2 s.

Commands from the repository root

Install the pinned dependencies using README.md and docs/commands.md. Training is optional for supplied-checkpoint playback.

```
export ISAAC_PYTHON=/path/to/isaac/environment/bin/python
./scripts/fetch_agibot_model.sh
./scripts/import_x2_isaac.sh
CKPT=reports/checkpoints/x2_supine_model3847.pt
./scripts/play_isaac.sh "$CKPT" --environment relaxed_v4 \
  --device cuda:0 --viz kit --start-delay 2 --seeds 101 \
  --output /tmp/x2_play.json
./scripts/evaluate_isaac.sh "$CKPT" --environment relaxed_v4 \
  --device cuda:0 --output /tmp/x2_evaluation.json
```

The two-second preview freezes the initial pose; physics and the episode clock do not advance. Seeds select small pose perturbations. Each new training run saves its checkpoint, reward PNG/CSV and manifest under its own timestamped logs/rsl_rl directory.

For ROS: build with scripts/build_ros.sh, start the Isaac policy server, launch both ROS nodes in a separately prepared terminal, then call /x2/start_recovery. Ordered commands and expected outputs are in docs/commands.md and docs/ros_live_validation.md.

Evidence and references

Evidence: reports/ (weights, training graph, evaluation and ROS logs); docs/parameter_provenance.md (parameter sources); docs/development_history.md (failures and fixes). Tested host: Ubuntu 22.04.5, RTX 5080 Laptop GPU with 16 GB.

- [1] HumanUP, RSS 2025. doi:10.15607/RSS.2025.XXI.063
- [2] HoST, RSS 2025. doi:10.15607/RSS.2025.XXI.064
- [3] FRASA, arXiv:2410.08655, Section IV-D
- [4] Proximal Policy Optimization, arXiv:1707.06347